



PRODUCT NUMBER

600 V, 52.14 A, 49 mΩ N-channel GaN Power MOSFET

1 Description

Our Mission We build fast, efficient, and reliable semiconductor devices specifically for high-performance switching applications.

Why Choose Helios? At Helios Power Devices, switching performance is our top priority. We engineer our MOSFETs to minimize switching losses and reduce gate charge so you get reliable operation at high power and high frequency. By focusing on these core metrics, we give system designers exactly what they need to build smaller, more efficient, and thermally stable power electronics.

Material Selection To achieve our performance targets, we built this device using Gallium Nitride (GaN). Standard silicon simply cannot keep up with the demands of modern power electronics. GaN offers an exceptionally high critical electric field and excellent electron mobility, allowing our transistor to handle massive voltage and current while switching at top speeds.

Geometry and Specifications Our physical design is tailored to balance high current capacity with rapid switching times. Based on our final computational models, the device features the following geometry and doping profiles:

- **Gate Length (L):** 1 μm to ensure a short transit time for high-frequency operation.
- **Gate Width (W):** 50 mm to support a high maximum drain current.
- **Oxide Thickness (t_{ox}):** 30 nm to maintain strong gate control.
- **Donor Doping (N_d):**
 - Contacts: 10^{19} cm^{-3}
 - Drift Region: $3.69 * 10^{18} \text{ cm}^{-3}$
- **Acceptor Doping (N_a):** $4.877 * 10^{16} \text{ cm}^{-3}$

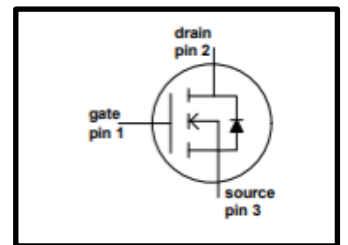
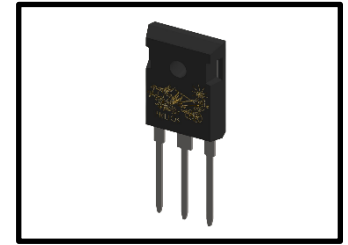
This specific combination of a wide channel, tight gate length, and optimized GaN doping allows the device to process high power loads without sacrificing the speed required for demanding RF amplifiers and power converters.

Features

- Enhancement-mode operation with a positive threshold voltage
- Low modeled on-resistance at the design point
- Fast switching with GHz-range transit frequency
- Strong transconductance for fast drive response

Applications

High-power switching stage, DC-DC converters, and fast control loops.



Key Performance Parameters

Parameter	Value	Unit
$V_{(\text{BR})\text{DSS}}$	600	V
$V_{\text{GS(th)}}$	1.71	V
$I_{\text{D,cont}}$	52.14	A
$R_{\text{DS(on)}}$	49	mΩ
P_{tot}	200	W

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2 Maximum Ratings

at $T_j = 25^\circ\text{C}$, unless otherwise specified.

Parameter	Symbol	Values			Unit	Condition / note
		Min.	Typ.	Max.		
Drain-source voltage	$V_{DS,max}$	-	-	500	V	-
Gate-source voltage	V_{GS}	3	10	15	V	Typ. is recommended
Continuous drain current	I_D	-	-	52.14	A	$T_C = 25^\circ\text{C}$, $V_{GS} = 10\text{ V}$
	I_D	-	-	30.90	A	$T_C = 125^\circ\text{C}$, $V_{GS} = 10\text{ V}$
Pulsed drain current	$I_{D,pulse}$	-	-	237	A	$T_C = 25^\circ\text{C}$
Power dissipation	P_{tot}	-	-	200	W	$T_C = 25^\circ\text{C}$
Operating / storage temp.	T_J, T_{stg}	-30	25	125	$^\circ\text{C}$	-
MOSFET dv/dt ruggedness	dv/dt	-	-	75.60	V/ns	-
Avalanche energy, single pulse	E_{AS}	-	-	0.46092	mJ	$I_D = 10\text{ A}$, $V_{DD} = 50\text{ V}$
Avalanche energy, repetitive	E_{AR}	-	-	0.071345	mJ	$I_D = 10\text{ A}$, $V_{DD} = 50\text{ V}$
Avalanche current, repetitive	I_{AR}	-	-	11.4367	A	-

3 Thermal Characteristics

Standards for the TO-247 Package

Parameter	Symbol	Values			Unit	Condition / note
		Min.	Typ.	Max.		
Thermal resistance, junction-case	R_{thJC}	-	-	0.50	$^\circ\text{C/W}$	-
Thermal resistance, junction-ambient	R_{thJA}	-	-	62	$^\circ\text{C/W}$	Leaded
Soldering temperature	T_{sld}	-	-	260	$^\circ\text{C}$	1.6 mm (0.063 in.) from case for 10 s

4 Electrical Characteristics

at $T_j = 25^\circ\text{C}$, unless otherwise specified.

Static characteristics

Parameter	Symbol	Values			Unit	Condition / note
		Min.	Typ.	Max.		
Gate threshold voltage	$V_{GS(th)}$	1.44	1.71	1.89	V	$V_{DS} = 500\text{ V}$; $I_D = 52.14\text{ A}$; $T_j = 25^\circ\text{C}$
Gate-source plateau voltage	$V_{GS(pl)}$	5.17	5.59	5.95	V	$V_{DS} = 500\text{ V}$; $I_D = 52.14\text{ A}$; $T_j = 25^\circ\text{C}$
Drain-source breakdown voltage	$V_{(BR)DSS}$	600	-	-	V	$V_{GS} = 0\text{ V}$
Drain-source on-state resistance	$R_{DS(on)}$	46	49	52	m Ω	$T_j = 25^\circ\text{C}$; $I_D = 52.14\text{ A}$
		125	140	149		$T_j = 125^\circ\text{C}$; $I_D = 30.90\text{ A}$
Gate Resistance	$R_{G,int}$	-	135	-	m Ω	Resistance ignores nonidealities
Transconductance	g_m	-	57.35	-	S (A/V)	-

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Dynamic characteristics

Parameter	Symbol	Values			Unit	Condition / note
		Min.	Typ.	Max.		
Input capacitance	C_{iss}	-	61.39	-	pF	$V_{DS} = 500\text{ V}; V_{GS} = 10\text{ V}$
Output capacitance	C_{oss}	-	12.09	-		
Reverse transfer capacitance	C_{rss}	-	11.51	-		
Coss stored energy	E_{oss}	-	1.51	-	μJ	$V_{DS} = 500\text{ V}; V_{GS} = 10\text{ V}$
Turn-on delay time	$t_{d(on)}$	-	12.6440	-	ns	$I_D = 52.14\text{ A}; V_{DS} = 500\text{ V}; V_{GS} = 10\text{ V}; R_G = 6\ \Omega$
Rise time	t_r	-	8.0630	-		
Turn-off delay time	$t_{d(off)}$	-	12.7893	-		
Fall time	t_f	-	6.6140	-		

5 Gate Charge Characteristics

Parameter	Symbol	Values			Unit	Condition / note
		Min.	Typ.	Max.		
Gate to source charge	Q_{gs}	-	0.3432	-	nC	$V_{GS} = 0\text{ V to }10\text{ V}; V_{DS} = 500\text{ V}; I_D = 52.14\text{ A}$
Gate to drain charge	Q_{gd}	-	5.7551	-		
Gate charge total	Q_g	-	6.0983	-		

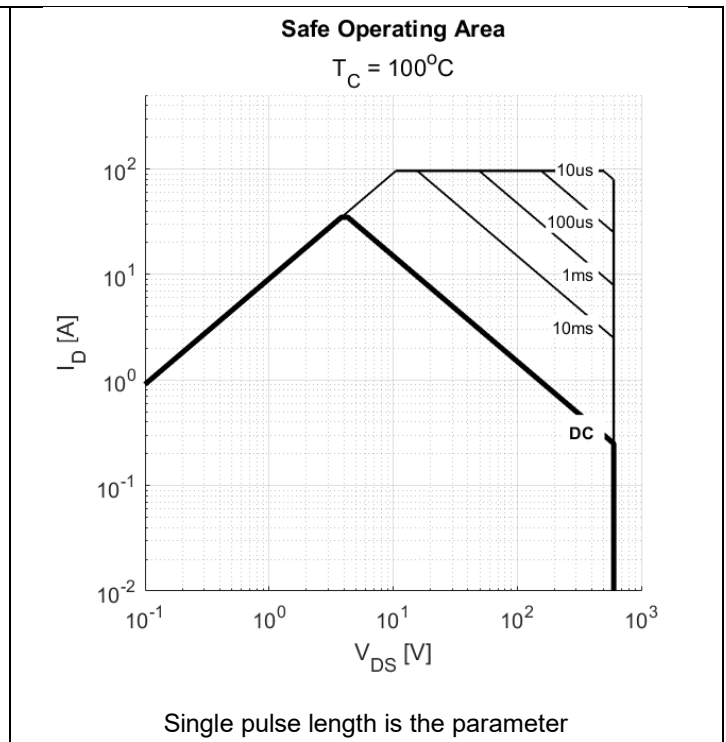
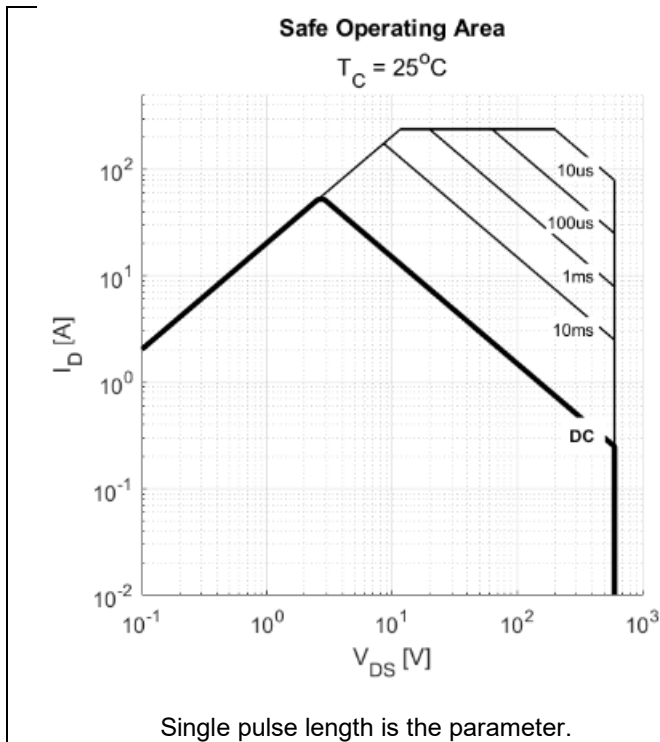
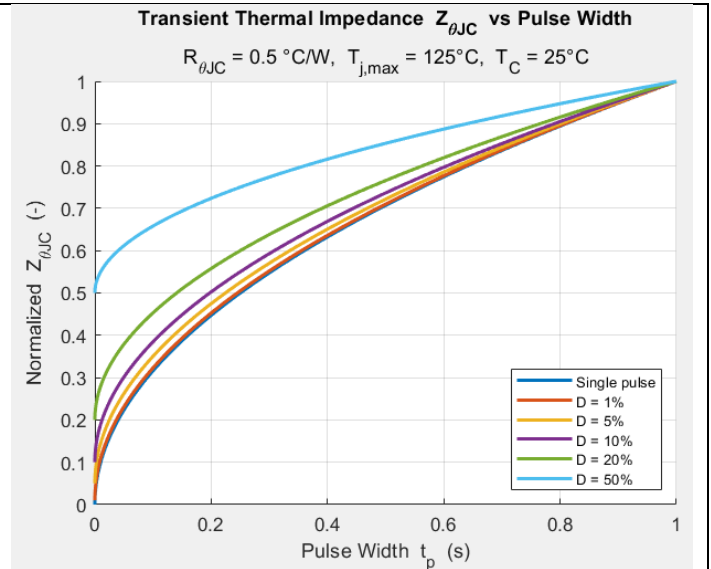
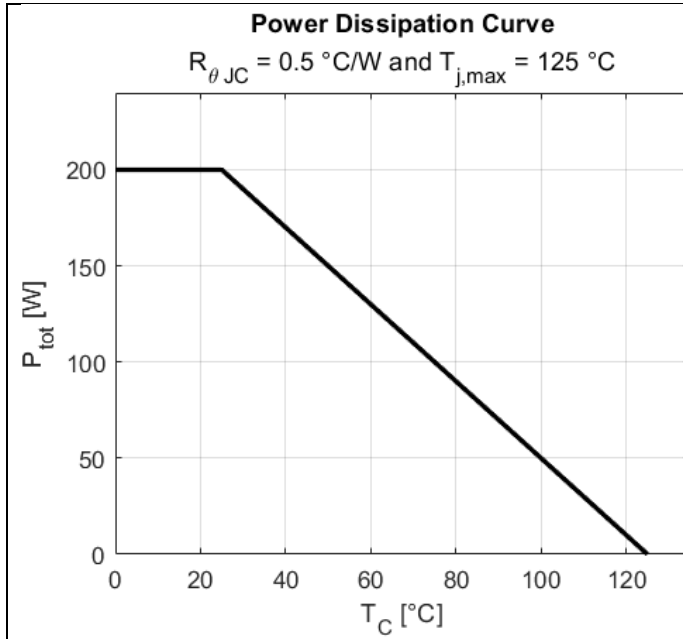
6 Reverse Diode Characteristics

Parameter	Symbol	Values			Unit	Condition / note
		Min.	Typ.	Max.		
Diode forward voltage	V_{SD}	-	3.271	-	V	$V_{GS} = 0\text{ V}; I_F = 26.3\text{ A}; T_j = 25\text{ }^\circ\text{C}$
Reverse recovery time	t_{rr}	-	19.19	-	ns	$V_R = 400\text{ V}, I_F = 26.3\text{ A}, di_F/dt = 100\text{ A}/\mu\text{s}$
Reverse recovery charge	Q_{rr}	-	18.41	-	nC	
Peak reverse recovery current	I_{mm}	-	1.919	-	A	

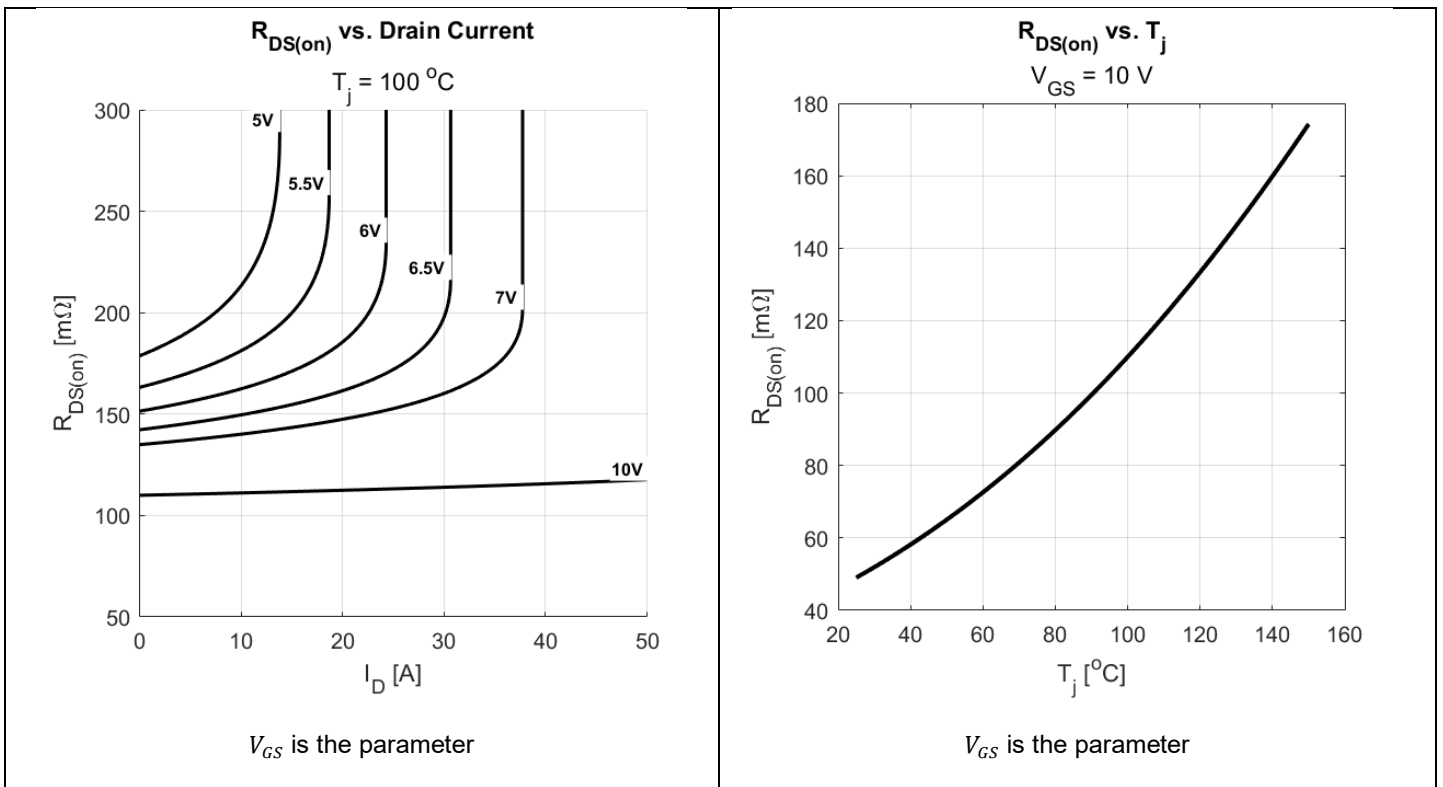
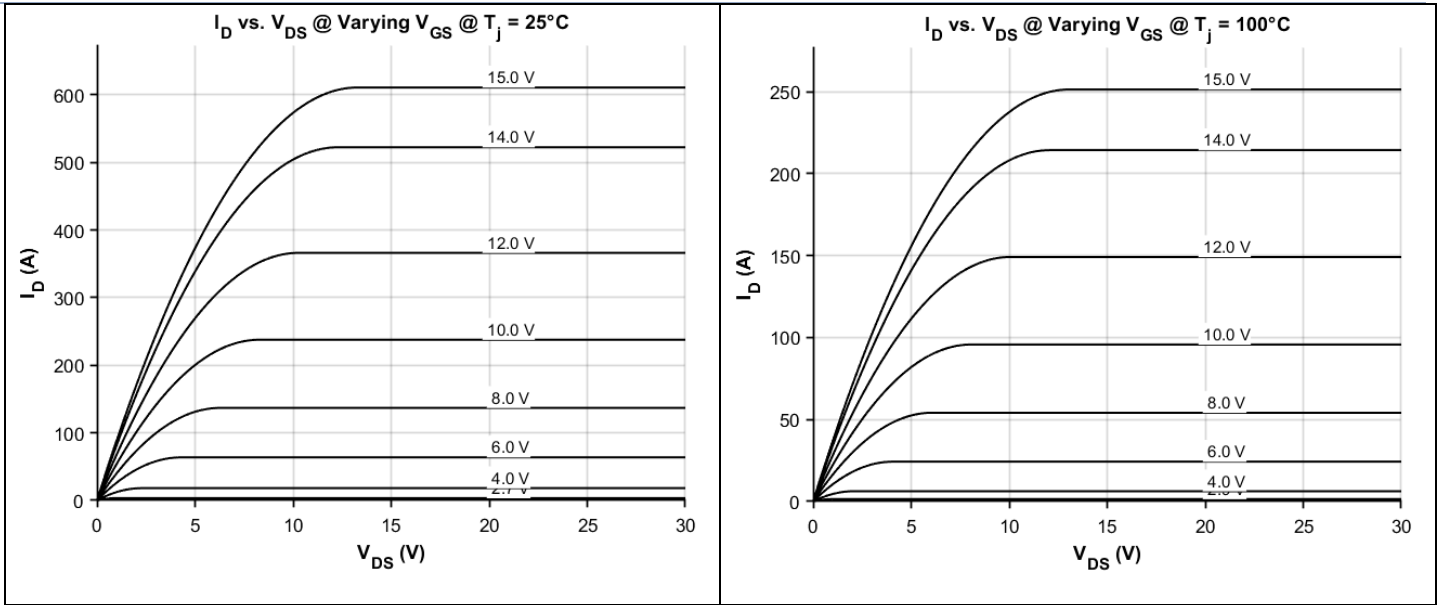
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5 Characteristics Diagrams

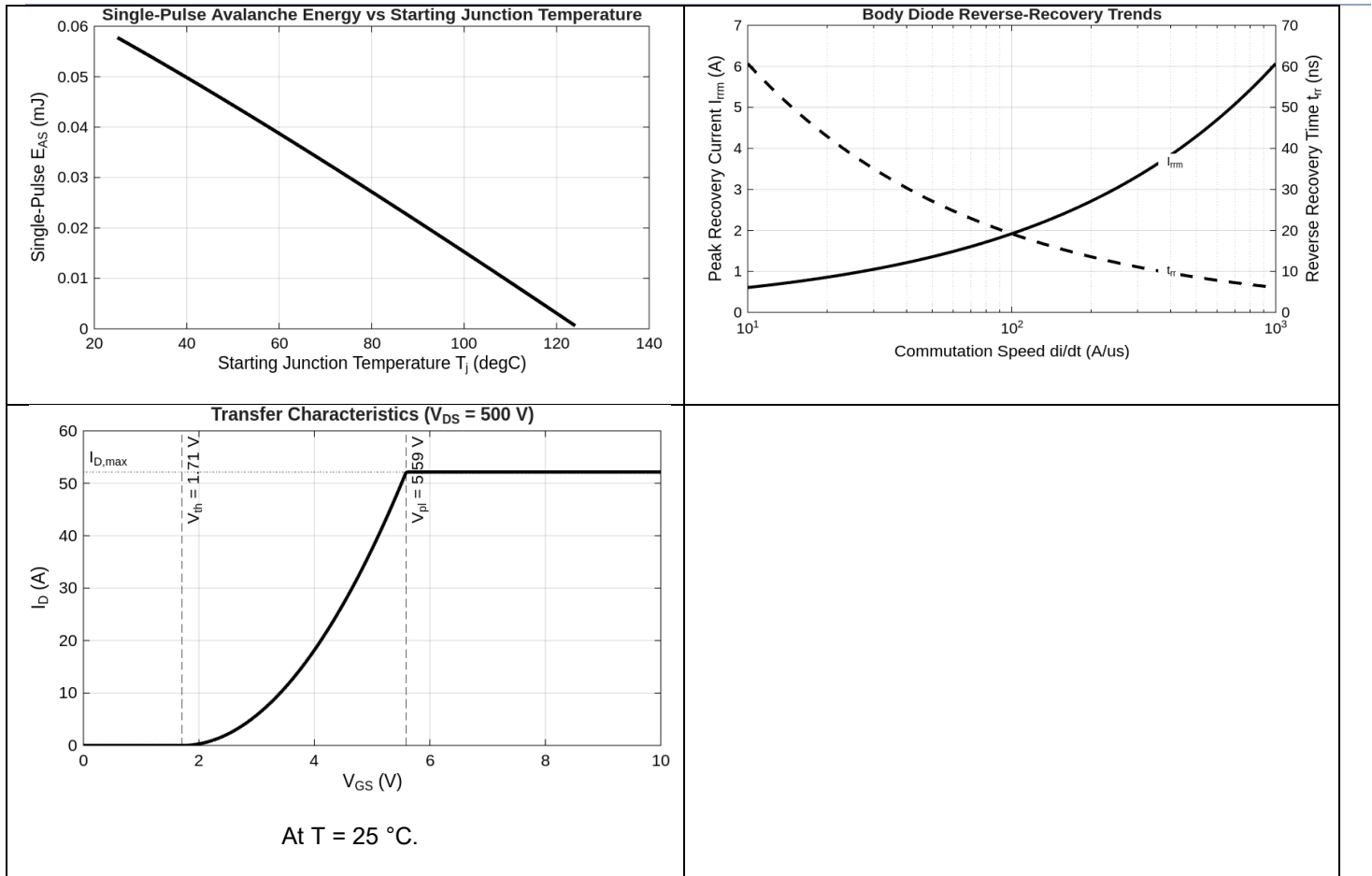
Plots generated from the MATLAB model.



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6 Package Outlines

